

HIGHVALE, AB, Canada

WMO: 712410

Lat: 53.449N

Lon: 114.464W

Elev: 747

StdP: 92.67

Time Zone: -7.00 (NAM)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.0	(c) -25.9	(d) -32.0	(e) 0.2	(f) -29.0	(g) -28.4	(h) 0.3	(i) -25.7	(j) 11.8	(k) 0.3	(l) 10.0	(m) -6.8	(n) 2.3	(o) 260	(p) 0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.5	(c) 27.6	(d) 18.5	(e) 25.8	(f) 17.5	(g) 24.1	(h) 16.5	(i) 20.0	(j) 25.4	(k) 18.9	(l) 24.1	(m) 17.8	(n) 22.5	(o) 3.3	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	14.3	22.7	17.0	13.2	21.2	15.9	12.4	20.1	60.7	25.2	56.8	24.2	53.2	22.5	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.1	(b) 9.4	(c) 8.0	(e) -33.5	(f) 30.9	(g) 3.7	(h) 2.0	(i) -36.2	(j) 32.3	(k) -38.4	(l) 33.5	(m) -40.4	(n) 34.6	(o) -43.1	(p) 36.0		
(4) 11.1	(5) 9.4	(6) 8.0	(7) -32.9	(8) 23.9	(9) 4.3	(10) 3.6	(11) -36.0	(12) 26.5	(13) -38.5	(14) 28.6	(15) -40.9	(16) 30.7	(17) -44.0	(18) 33.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-11.0	-6.4	-3.1	5.3	10.6	14.7	17.4	16.3	11.7	5.3	-2.7	-7.6
	DBStd	11.41	9.47	7.65	7.99	5.40	4.49	3.05	2.96	3.36	3.97	4.76	6.46	8.75
	HDD10.0	2833	651	459	405	154	47	2	1	2	28	157	382	545
	HDD18.3	5186	909	692	663	391	242	115	52	79	201	406	632	803
	CDD10.0	737	0	0	0	13	64	142	231	199	78	10	1	0
	CDD18.3	48	0	0	0	0	1	5	24	17	1	0	0	0
	CDH23.3	530	0	0	0	1	26	74	233	170	26	2	0	0
	CDH26.7	99	0	0	0	0	3	13	49	32	2	0	0	0
Wind	WSAvg	3.4	2.9	3.3	3.7	3.8	3.8	3.6	3.3	3.0	3.4	3.6	3.4	3.6
Precipitation	PrecAvg	491	25	16	19	31	55	87	94	63	39	24	21	17
	PrecMax	652	75	30	36	55	110	175	208	138	87	76	61	47
	PrecMin	321	2	1	3	5	18	25	46	31	9	5	4	5
	PrecStd	79	17	8	8	15	23	34	39	21	22	17	15	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	9.9	14.5	22.0	26.9	28.4	30.8	30.1	26.3	21.5	13.1	11.6
		MCWB	4.0	4.2	6.4	11.0	13.1	17.5	19.6	21.5	16.2	13.1	6.5	4.6
	2%	DB	6.2	7.3	11.1	19.1	23.6	25.3	27.8	27.1	23.4	17.1	9.6	8.1
		MCWB	2.7	2.6	4.8	9.6	13.1	15.4	18.9	19.6	15.2	10.3	5.3	3.5
	5%	DB	3.9	5.1	8.9	16.5	21.4	23.1	25.9	25.2	21.1	14.3	7.1	5.9
		MCWB	1.1	1.6	3.7	8.0	12.1	14.2	18.1	18.1	14.3	8.7	3.5	2.5
	10%	DB	1.7	3.2	6.6	13.8	19.0	21.1	24.1	23.3	18.7	12.1	4.7	3.7
		MCWB	-0.2	0.2	2.6	6.7	10.7	13.7	17.3	17.6	13.4	7.6	1.9	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.7	7.5	11.9	15.6	18.7	21.3	22.2	18.4	14.0	7.6	5.6
		MCDB	8.0	9.2	13.5	19.9	23.6	25.4	26.9	27.0	22.9	20.7	12.3	10.3
	2%	WB	3.0	2.9	5.4	10.2	14.2	17.1	20.0	20.2	16.6	11.1	5.5	3.8
		MCDB	5.8	6.7	10.1	18.0	20.9	23.0	25.7	25.5	21.7	16.2	9.0	7.3
	5%	WB	1.6	1.7	4.0	8.6	13.0	15.9	19.0	19.1	15.1	9.5	4.0	2.5
		MCDB	3.8	4.7	8.1	15.4	19.5	21.1	24.8	23.8	19.7	13.4	6.9	5.6
	10%	WB	-0.2	0.4	2.8	7.1	11.8	14.9	17.9	17.9	13.7	8.1	2.3	1.0
		MCDB	1.8	3.0	6.3	12.9	17.7	19.2	23.0	22.2	18.1	11.6	4.3	3.4
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.7	9.6	11.0	11.0	10.0	10.5	10.8	10.7	9.4	7.8	8.5
		MCDBR	10.2	12.0	12.3	15.1	14.7	14.0	13.6	14.0	15.2	12.9	9.9	8.9
		MCWBR	7.6	8.5	7.5	7.7	6.8	6.2	6.4	7.4	8.2	7.4	6.4	6.1
	5% WB	MCDBR	11.2	11.6	11.9	14.2	13.0	12.0	12.9	13.3	14.0	12.7	9.9	9.4
		MCWBR	8.9	8.5	7.6	7.8	6.3	6.2	6.8	7.8	8.0	7.8	6.8	6.8
Clear-Sky Solar Irradiance	taub	0.249	0.261	0.295	0.332	0.346	0.354	0.355	0.361	0.314	0.283	0.256	0.240	
	taud	2.346	2.385	2.392	2.379	2.382	2.375	2.426	2.362	2.494	2.560	2.464	2.319	
	Ebn at Noon	754	863	898	901	902	895	887	860	869	833	749	688	
	Edh at Noon	55	73	91	106	112	114	106	107	82	62	49	46	
All-Sky Solar Radiation	RadAvg	0.96	1.98	3.45	4.63	5.56	5.93	6.06	5.01	3.48	2.00	1.00	0.69	
	RadStd	0.07	0.15	0.22	0.42	0.40	0.40	0.37	0.34	0.36	0.21	0.09	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page